

ANONYMOUS EXPOSES CHINESE HACKERS

Security firm Mandiant were successfully able to name two Chinese hackers, who they claim to steal corporate data from American firms, through the accidental help of Anonymous.
<http://dgit.in/I2Ma0A5>

THE MAGIC OF GIFS

GIFs have come of age. They are no longer few

frames of animated nonsense, increasingly being used as instruments of storytelling. And the trend is steadily growing, as everyone has a story to tell.
<http://dgit.in/Y7f4v9>

7 REASONS TO SWITCH TO OUTLOOK.COM

Hotmail is dead, long live Hotmail!

As Microsoft migrates millions of Hotmail accounts over to its latest Outlook.com email service, here's a strong case as to why you should switch to Hotmail's successor.
<http://dgit.in/YzUUYw>

BILL GATES QUIZZED ONLINE

Reddit's immensely popular AMA or

Ask Me Anything hosted the chairman and co-founder of Microsoft, Bill Gates. And he fielded questions from random Reddit users. The resulting "interview" is worth a read.
<http://dgit.in/Y7pvie>

Human noses: Quantum smelling devices

- By John Hewitt



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Technology has given mankind a machine alternative that bests every sense except one — smell. One man, a perfumer and biophysicist named Luca Turin, has over the last two decades pieced together a highly-contested theory that seeks to explain how odors are detected and transduced — converted from molecules into neural spikes. This week he published incontrovertible evidence that may silence his many critics once and for all.

Arguments against the theory hinge on a particular molecule known as carvone. As shown in the picture to the right, carvone comes in two enantiomers — in other words, flavors — that are mirror images of each other. Inexplicably, one enantiomer — the R- form — smells of spearmint, while the S- form smells of caraway or dill.

The traditional lock and key theory of smell holds that odorants bind into matched receptor pockets that detect shapes. This theory works fine for carvone so long as there are different receptors for each version. The main limitation of the lock and key concept is that there are many examples where molecules with completely different shapes smell similarly. For example, borane, which lacks any sulfur bonds, still has the rotten egg smell normally associated to those bonds. It turns out that borane has similar peaks in its infrared (IR) or vibrational spectrum that overlap those of sulfur bonds. Turin maintains that when

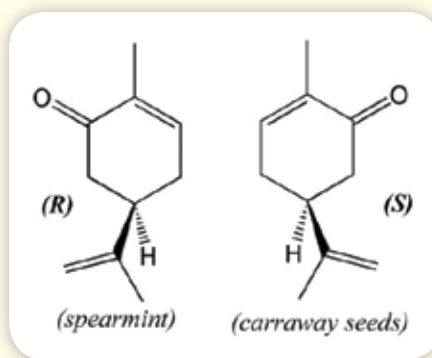
an odorant binds into the right pocket, electrons can tunnel through it in a way that depends on the frequency at which the molecule intrinsically vibrates. The entire vibrational spectrum would be divided up with each receptor commanding a small chunk of it. In this interpretation, the nose becomes more-or-less a chemical spectroscopist.

The problem carvone presents to Turin's theory is that although the mirror images smell different, they possess identical IR spectrums. In dramatic counterpoint to this seeming roadblock, Turin added butanol to a sample of dill carvone and made it smell like spearmint. Butanol has the same kind of bond found in carvone but it is small enough to slip into receptor pockets along with it. The critics shifted gears by demonstrating that the small molecule acetophenone

already showed in *Drosophila*, the fruit fly, that the deuterated forms of acetophenone could be discriminated at least behaviorally, but that is just not as experimentally convincing as a human directly reporting what they smell.

Turin's masterstroke, published recently in *Plos One*, demonstrates that one grenade does not bring down the whole fort. Turin hypothesized that a larger molecule like a musk, with more points that could be deuterated, would be more detectable to humans. Musk molecules are about as large as you can get as far as being able to fit inside olfactory receptors. At a weight of nearly 300Da (daltons), most of them can hardly be considered volatile and tend to reside on the ground or stick to objects like trees. Just a couple molecules of this potent and expensive stuff is, at least for the deer or wild boar, plenty enough to make their world go round.

As old theory yields to hard won data, machine sensing will be the immediate benefactor, while, for us, the artificial nose, enhanced perception, and greater understanding of how we internally package the larger realities of our sensory landscape may soon follow. Fully decoding this mysterious prowess remains the final frontier of sensory science. The genome is littered with countless defunct receptor proteins cast aside in recent evolution along with our atrophied primate olfactory bulbs. The olfactory organ is a privileged area of the brain where new neurons continually migrate and interconnect. Eventual redeployment of the lost intuitions of the canine through proper technology will enrich our human experience in ways we can only now imagine.



smells the same to humans even when its vibrational spectrum is altered by substituting some of its hydrogens with heavier deuterium atoms. This substitution causes the bonds of the molecule to oscillate more slowly, effectively altering the global undulation of the entire molecule and the ability of electrons to tunnel through it. Turin had